

CERTIFIED ETHICAL HACKER

Master Program |
CEH AI v13

Course Objective

The Masterclass Ethical Hacking – CEH v13 program from eHack Academy is a 4-month intensive cybersecurity course focused on preparing you for the Certified Ethical Hacker (CEH) v13 certification and building practical ethical hacking skills. It teaches how to think like a hacker in order to identify and exploit vulnerabilities across networks, systems, and applications using real-world tools and methodologies, with an emphasis on hands-on labs and AI-enhanced learning. The curriculum covers essential ethical hacking domains such as reconnaissance, scanning, exploitation, social engineering, and defensive countermeasures to equip learners for roles in penetration testing and cybersecurity defence.

Earn 3 Global Certifications 🏆



CEH CEH PRACTICAL CEH MASTER

4 MONTHS
Duration

100+
Training Hours

6 MONTHS
Membership

Course Syllabus

Module 01

Introduction to Ethical Hacking -

Covers the fundamentals of ethical hacking, information security controls, relevant laws, and standard procedures.

What you'll learn:

- Information Security Fundamentals
- Ethical Hacking Concepts
- Information Security Controls
- Information Security Laws and Standards
- Cyber Kill Chain Methodology

Module 02

Footprinting and Reconnaissance -

Acquire techniques and tools for gathering data before an attack, a critical pre-attack phase.

What you'll learn:

- Footprinting Concepts and Objectives
- Footprinting through Search Engines
- Footprinting through Web Services
- Footprinting through Social Networking Sites
- Website Footprinting and Email Footprinting

Module 03

Scanning Networks -

Focuses on various network scanning techniques and associated countermeasures.

What you'll learn:

- Network Scanning Concepts
- Scanning Tools and Techniques
- Host Discovery
- Port and Service Discovery
- Network Scanning Countermeasures

Module 04

Enumeration -

Teaches methods for extracting system information for penetration testing purposes.

What you'll learn:

- Enumeration Concepts
- NetBIOS Enumeration
- SNMP Enumeration
- LDAP Enumeration
- NTP and NFS Enumeration

Module 05

Vulnerability Analysis -

Involves identifying security weaknesses within systems.

What you'll learn:

- Vulnerability Assessment Concepts
- Vulnerability Assessment Solutions
- Vulnerability Scoring Systems
- Vulnerability Assessment Reports
- Vulnerability Management Life Cycle

Module 06

System Hacking -

Covers exploiting and protecting operating systems.

What you'll learn:

- System Hacking Concepts
- Password Cracking Techniques
- Escalating Privileges
- Executing Applications and Hiding Files
- Clearing Logs and Covering Tracks

Module 07

Malware Threats -

Explores different types of malware, such as viruses, Trojans, and ransomware.

What you'll learn:

- Malware Concepts

- APT and Fileless Malware
- Trojan Horses
- Viruses and Worms
- Malware Analysis and Countermeasures

Module 08

Sniffing -

Focuses on capturing and analyzing network traffic securely.

What you'll learn:

- Sniffing Concepts
- Sniffing Techniques
- Sniffing Tools
- MAC Flooding and ARP Poisoning
- Sniffing Countermeasures

Module 09

Social Engineering -

Delves into manipulation tactics and prevention strategies.

What you'll learn:

- Social Engineering Concepts
- Social Engineering Techniques
- Insider Threats and Identity Theft
- Phishing and Impersonation
- Social Engineering Countermeasures

Module 10

Denial-of-Service -

Explains attack methods and defense strategies against Denial-of-Service attacks.

What you'll learn:

- DoS/DDoS Concepts
- DoS/DDoS Attack Techniques
- Botnets and DDoS Attack Tools
- DoS/DDoS Case Studies
- DoS/DDoS Countermeasures

Module 11

Session Hijacking -

Covers understanding session-based attack scenarios.

What you'll learn:

- Session Hijacking Concepts
- Application Level Session Hijacking
- Network Level Session Hijacking
- Session Hijacking Tools
- Session Hijacking Countermeasures

Module 12

Evading IDS, Firewalls, and Honeypots -

Teaches techniques for bypassing detection systems safely.

What you'll learn:

- IDS, Firewall, and Honeypot Concepts
- IDS/Firewall Evasion Techniques
- Firewall and IDS Evasion Tools
- Detecting Honeypots
- IDS/Firewall Evasion Countermeasures

Module 13

Hacking Web Servers -

Focuses on securing and testing web infrastructure.

What you'll learn:

- Web Server Concepts
- Web Server Attacks
- Web Server Attack Methodology
- Web Server Attack Tools
- Web Server Security and Countermeasures

Module 14

Hacking Web Applications -

Involves analyzing application vulnerabilities.

What you'll learn:

- Web Application Concepts
- Web Application Threats

- Web Application Hacking Methodology
- Web API and Webhook Hacking
- Web Application Security Testing

Module 15

SQL Injection -

Covers detecting and preventing database attacks through SQL injection.

What you'll learn:

- SQL Injection Concepts
- Types of SQL Injection
- SQL Injection Methodology
- SQL Injection Tools
- SQL Injection Countermeasures

Module 16

Hacking Wireless Networks -

Addresses Wi-Fi security and encryption flaws.

What you'll learn:

- Wireless Concepts and Encryption
- Wireless Threats and Attacks
- Wireless Hacking Methodology
- Wireless Hacking Tools
- Wireless Security and Countermeasures

Module 17

Hacking Mobile Platforms -

Explores mobile app vulnerabilities and security for platforms like Android and iOS.

What you'll learn:

- Mobile Platform Attack Vectors
- Hacking Android Platforms
- Hacking iOS Platforms
- Mobile Device Management
- Mobile Security Guidelines and Tools

Module 18

IoT and OT Hacking -

Focuses on securing Internet of Things (IoT) and Operational Technology (OT) devices and systems.

What you'll learn:

- IoT Hacking Concepts
- IoT Threats and Attacks
- OT Concepts and Attacks
- IoT and OT Hacking Methodology
- IoT and OT Security Countermeasures

Module 19

Cloud Computing -

Covers cloud computing concepts, threats, security, and AI-driven container technology, along with serverless computing security measures.

What you'll learn:

- Cloud Computing Concepts
- Container Technology
- Serverless Computing
- Cloud Computing Threats and Attacks
- Cloud Security and Countermeasures

Module 20

Cryptography -

Covers encryption algorithms, Public Key Infrastructure (PKI), cryptographic attacks, and cryptanalysis.

What you'll learn:

- Cryptography Concepts
- Encryption Algorithms
- Public Key Infrastructure (PKI)
- Cryptographic Attacks
- Cryptanalysis and Countermeasures